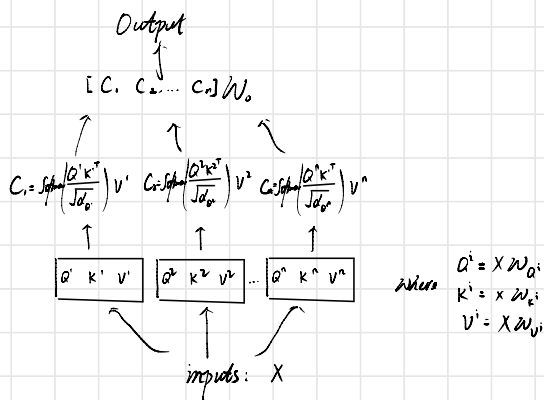


## Standard multi-head attention



problem of multi-head attention: the KV Cache bottleneck

at each head, the key and query of the previous tokens will be cached, to avoid recalculating, however they get longer and longer.

Solution:

Grouped Query attention: The Query heads are divided into groups. each group of query heads shares a single Key and value head.

